Constraints on the ^{7}Be$(p,\gamma)^8$B radiative capture rate from charge symmetry

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Cross sections for the dipole radiative capture reactions ^{7}Li$(n,\gamma)^8$Li and ^{7}Be$(p,\gamma)^8$B are calculated in a two-body model, which is based on a Woods-Saxon parametrization of the nuclear interaction. The well depth is adjusted for each reaction channel so that the measured separation energies and *s*-wave scattering lengths are reproduced. The calculations are repeated for a wide range of the radius and the diffuseness of the interaction. The predicted *S* factor for the radiative proton capture on ^{7}Be falls within a surprisingly narrow range of values when the model is calibrated to reproduce measurements of the mirror reaction ^{7}Li$(n,\gamma)^8$Li. The simplified model used here is consistent with the shell model approach for proton capture on ^{7}Be but it gives a significantly smaller cross section for neutron capture on ^{7}Li.

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Much effort has gone into determining the radiative proton capture rate on ^{7}Be at energies relevant to the interior of the Sun. Measurements are difficult and it is necessary to make extrapolations to low energies. As a consequence, the value of the *S* factor that was recommended in 1998 [1] had a 10–20 % uncertainty, $S_{17}(0)=19\pm^4_2$ eV b. The uncertainty has been reduced to less than 4% in recent capture measurements [2,3], and the extrapolated *S* factor has increased to $21-22$ eV b. Values extracted from Coulomb dissociations experiments [4–6] are smaller $(18-19$ eV b$)$ but the uncertainty is larger.

An alternative method is to use information about the interaction of low-energy neutrons with ^{7}Li and combine it with charge symmetry to predict the capture rate of low-energy protons on ^{7}Be. An example of this approach is Ref. [7], where the neutron transfer to ^{7}Li was measured and used to infer the *S* factor for ^{8}B. In this work information about the low-energy neutron scattering [8] and radiative capture [9] on ^{7}Li will be used, together with charge symmetry, to predict the *S* factor for ^{8}B. This requires a model for calculating the bound state and the scattering states of the nucleon being captured.

A simple and commonly used approach is to adopt a two-body model. The model that will be used here is based on a Woods-Saxon parametrization of the nuclear interaction between the valence nucleon and the core:

$$V_N(r) = \left[1 - F_{\text{s.o.}}(\mathbf{l}\cdot\mathbf{s})\frac{r_0}{r}\frac{d}{dr}\right]\frac{V_0(lj,J)}{1+\exp[(r-R)/a]}. \qquad (1)$$

The spin-orbit strength $F_{\text{s.o.}}$ is kept fixed at 0.351 fm [10]. The nuclear radius is parametrized as $R=r_0(A-1)^{1/3}$ with $A=8$. The core nucleus is assumed to be inert with a ground-state spin of $3/2^-$. The orbital and total angular momenta of the valence particle are denoted by (lj). The total channel spin *J* is obtained by coupling the $3/2^-$ spin of the core to the angular momentum *j* of the valence nucleon. The well depth $V_0(lj,J)$ is adjusted for each reaction channel. The interaction between the valence proton and the ^{7}Be core is supplemented with the Coulomb interaction, which is calculated in the sharp-sphere approximation for the charge density of the core, with charge radius $R_C=1.25(A-1)^{1/3}$.

The 2^+ ground state of ^{8}B or ^{8}Li is described in terms of a pure $p_{3/2}$ orbit of the valence nucleon, and the well depth is adjusted so that the known separation energy ($S_p=0.1375$ MeV or $S_n=2.033$ MeV) is reproduced. The 1^+ channel is also described in terms of a $p_{3/2}$ orbit. It has a resonance at 0.637 MeV in ^{8}B and a bound state at -1.052 MeV in ^{8}Li, and the well depths are adjusted accordingly. In the shell model approach [11,12], on the other hand, one uses a mixed $p_{3/2}+p_{1/2}$ ground-state wave function and we shall explore what difference that makes.

The low-energy radiative capture is dominated by *E*1 transitions. It can proceed from continuum *s* and *d* waves, so we need to calibrate the nuclear interaction in these channels. There are two *s*-wave channels, with $J^\pi=1^-$ and 2^-. The well depths $V(s_{1/2},1^-)$ and $V(s_{1/2},2^-)$ are adjusted so that the measured scattering lengths for neutrons on ^{7}Li—namely, $a(1^-)=0.87\pm0.07$ fm and $a(2^-)=-3.63\pm0.05$ fm [8]—are reproduced. The same well depths will be used for the proton *s*-wave scattering on ^{7}Be. The calibration for *d* waves is more uncertain. Fortunately, this turns out not to be so critical for the calculation of the low-energy radiative capture. The well depth for *d* waves is therefore set equal to the value of $V(s_{1/2},1^-)$, except when otherwise stated.

Having calibrated the two-body interactions (1), the radiative *E*1 capture cross sections can now be calculated as described, for example, in Ref. [10]. The results for ^{8}B will be presented in terms of the *S* factor extrapolated to zero energy, $S_{17}(0)$. The results for ^{8}Li are presented in terms of the ratio N_{nc} of the measured and the calculated cross sections for the radiative capture of thermal neutrons on ^{7}Li to the 2^+ ground state of ^{8}Li:

$$N_{nc} = \left.\frac{\sigma_{expt}(2^+)}{\sigma_{calc}(2^+)}\right|_{E_n^{th}}. \qquad (2)$$

The measured cross section is $\sigma_{expt}(2^+,E_n^{th})=40.56$ mb (±3 mb), where $E_n^{th}=0.025$ eV [9]. The correlation between $S_{17}(0)$ and N_{nc} will be discussed later on.

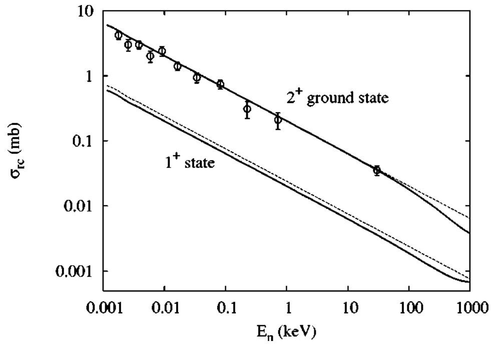

FIG. 1. Cross sections for radiative neutron capture on ^{7}Li to the 2^+ ground state ^{8}Li (upper curves and data) and to the 1^+ excited state (lower curves). The data are from Refs. [14,15]. The dashed curves show the $1/\sqrt{E_n}$ extrapolation of the thermal neutron cross sections [9]. The solid curves are the calculated cross sections discussed in the text.

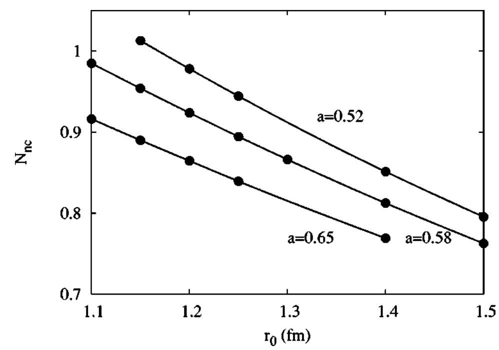

FIG. 2. The normalization factor defined in Eq. (2) for the radiative neutron capture on ^{7}Li is shown as a function of the nuclear radius parameter r_0 for three choices of the diffuseness a.

The results of the calculations will be interpreted in terms of the asymptotic normalization constant (ANC) [13]. The ANC of the ground-state wave function $\Psi_{2^+}(r)$ is denoted by $C(2^+)$. It is determined at large radial separations by the asymptotic form

$$\Psi_{2^+}(r) = \frac{C(2^+)}{r} W_{-\eta,l+1/2}(2\kappa r) \tag{3}$$

in terms of the Whittaker function $W_{-\eta,l+1/2}(2\kappa r)$, which depends on the Sommerfeld parameter η, the orbital angular momentum l, and the wave number κ.

The cross sections for the direct radiative neutron capture on ^{7}Li to the 2^+ ground state and to the 1^+ excited state of ^{8}Li are shown in Fig. 1. The parameters of the potential chosen here are $r_0=1.30$ and $a=0.58$ fm. The calculated cross sections (solid curves) have been multiplied by the factor $N_{nc}=0.866$, so that the measured thermal neutron cross section for the capture to the 2^+ ground state is reproduced. The calculated cross section for the capture to the 1^+ excited state has, in addition, been multiplied by the ratio of the measured spectroscopic factors of the two states [16], which is $S(1^+)/S(2^+)=0.48/0.87$. In this way a branching ratio of $\sigma(1^+)/\sigma(2^+)=0.10$ is predicted for thermal neutrons. This ratio is insensitive to the parameters (r_0,a) of the interaction, and it is in reasonable agreement with the measured value [9] of 0.12 ± 0.01.

The calculated capture cross sections follow a $1/\sqrt{E_n}$ dependence as expected [15]. The thin dashed lines in Fig. 1 show such an extrapolation of the thermal cross sections. Measured cross sections [14,15] for direct capture to the 2^+ ground state are also shown. They are consistent with the extrapolation. Data obtained at higher energies (see references in [15]) are not shown because they are influenced by the 3^+ resonance at 222 keV.

The normalization factors N_{nc} that have been obtained are shown in Fig. 2. The solid lines are interpolations between the calculated results, which are indicated by solid points. If the radiative neutron capture were determined entirely by the tail of the ground-state wave function, then one would expect according to Ref. [13] that the calculated cross section,—or, equivalently, N_{nc}^{-1}—would be proportional to the square of the ANC. From the calculations one obtains the relation

$$N_{nc}|C(2^+,{}^8\mathrm{Li})|^2 \approx 0.475 \pm 0.025\ (\mathrm{fm}^{-1}). \tag{4}$$

The uncertainty reflects the sensitivity to the parameters of the nuclear potential. Note that Eq. (4) is consistent with the square of the ANC extracted from transfer measurements [7], which is 0.432 ± 0.043 (fm^{-1}).

An interesting feature of the calculations is that the neutron capture from d waves plays an insignificant role below 100 keV. The s-wave capture is dominated by the 2^- channel, which provides about 74% of the total capture cross section at low energy. The reason is that the nuclear interactions in the two s-wave channels have been chosen differently, in order to reproduce the two measured scattering lengths. The radiative capture from the two s-wave channels would have been the same if the nuclear interactions were chosen to be identical.

The calculated S factor for the radiative proton capture on ^{7}Be is shown in Fig. 3. The parameter set used here, $r_0=1.30$ and $a=0.58$ fm, is the same that was used to produce

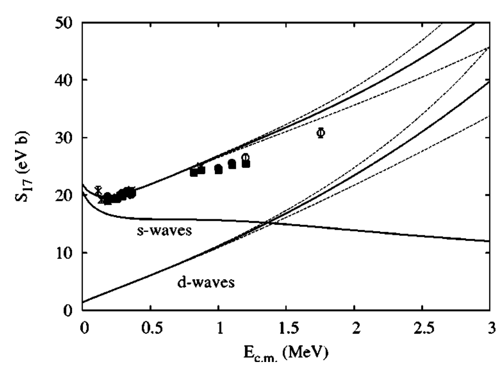

FIG. 3. S factor for the radiative proton capture on ^{7}Be as function of the c.m. energy. The data are from Ref. [2]. The solid curves show the separate contributions from s and d waves, and the sum. The potential parameters are $r_0=1.3$ and $a=0.58$ fm, and the spectroscopic factor is 1. The dashed curves were obtained for a d-wave well depth set equal to zero (lower dashed curves) or $V(s_{1/2},2^-)$ (upper dashed curves).

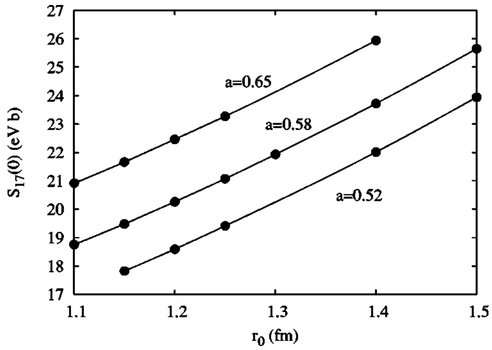

FIG. 4. S factor for radiative proton capture on ^{7}Be at zero energy, calculated as a function of the radius parameter r_0 for three values of the diffuseness a. The spectroscopic factor was set equal to 1.

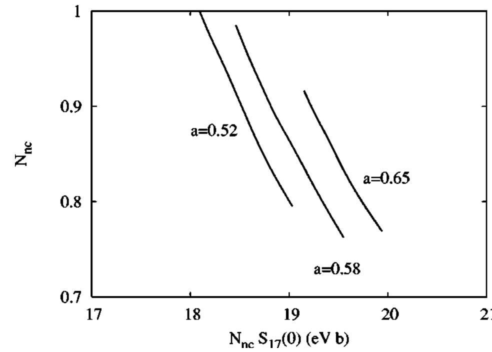

FIG. 5. Correlation between the normalization factor N_{nc} for neutron capture on ^{7}Li and the effective S factor, $N_{nc}\times S_{17}(0)$, for proton capture on ^{7}Be obtained for three values of the diffuseness.

Fig. 1. The spectroscopic factor of the ground state was assumed to be 1.0. The three solid curves show the separate contributions from the s- and d-wave capture, and the sum of these. The dashed curves illustrate the sensitivity to the choice of the well depth V_{dw} for d waves. The lower dashed curves were obtained by setting $V_{dw}=0$. The upper dashed curves were based on $V_{dw}=V_0(s_{1/2},2^-)$, whereas the solid curves were based on the standard choice $V_{dw}=V_0(s_{1/2},1^-)$. It is seen that the uncertainty in the d-wave capture does not play any role below 1 MeV but it is quite large above 2 MeV.

The contributions to the capture rate from the two s-wave channels turn out to be essentially identical below 1 keV. The intuitive explanation is that the Coulomb field from ^{7}Be effectively shields the interior of the nucleus from the low-energy protons. The same argument also explains why the low-energy d-wave capture is insensitive to the nuclear interaction strength V_{dw}.

The S factors that have been obtained at zero energy are shown in Fig. 4. The contribution from d waves is remarkably constant; it is about 6.1% in all of the cases shown in Fig. 4. From the calculations one obtains

$$S_{17}(0) = [38.7 \text{ (eV b fm)}]|C(2^+,{}^8\text{B})|^2. \tag{5}$$

This relation has previously been pointed out in Ref. [13]. It is very accurate in contrast to the approximate relation (4) for neutron capture. It appears that the radiative proton capture mainly probes the tail of the ground-state wave function, whereas the neutron capture also probes the interior part.

Assuming charge symmetry one would expect that the normalization factor N_{nc} obtained for the neutron capture should also be applied to the proton capture calculation. We can therefore estimate an effective S factor at zero energy by the product $N_{nc}\times S_{17}(0)$. The correlation between N_{nc} and the effective S factor that one obtains for each parameter set (r_0,a) of the nuclear interaction is shown in Fig. 5. It is seen that there is a rather tight constraint on the effective S factor. Another way to demonstrate this point is to combine the two expressions, Eqs. (4) and (5), which gives

$$N_{nc}S_{17}(0) = [18.5 \pm 0.9 \text{ (eV b)}]\frac{|C(2^+,{}^8\text{B})|^2}{|C(2^+,{}^8\text{Li})|^2}. \tag{6}$$

Here the ratio of the squares of the two ANCs was found to be 1.03 ± 0.02 (Ref. [7] used 1.055 ± 0.02), so we obtain N_{nc} $S_{17}(0)\approx 19.0\pm1.0$ eV b. This prediction is not unreasonable in comparison to the broad range of experimental values. Let us now compare to predictions of other models.

Descouvemont and Baye [17] have applied their three-body cluster model to study the two radiative capture reactions discussed above. They used a series of nucleon-nucleon interactions. The interaction that provided the best result for the low-energy neutron capture on ^{7}Li resulted in the S factor $S_{17}(0)=27$ eV b for ^{8}B. The authors claimed that the value of $S_{17}(0)$ cannot be significantly less than 24 eV b. This prediction is at the higher end of the experimental results.

In calculations based on the shell model [11,12] it has not been possible to reproduce simultaneously the low-energy data for the radiative neutron capture on ^{7}Li and the S factor for ^{8}B. Choosing a reasonable potential model, Barker [11] obtained what seemed like a reasonable S factor for ^{8}B, but the calculated neutron capture cross section was too large. Conversely, adjusting the model to reproduce the thermal neutron capture data, he obtained a small S factor $S_{17}(0)$ $\leqslant 16$ eV b. This is at the lower end of the values extracted from experiments. A similar problem was recognized in Ref. [12].

With such a wide range of model predictions it is of interest to understand what causes the differences. Let us take a closer look at the shell model approach [11,12]. Here the spin $1/2$ of the captured nucleon is coupled to the spin state $I_c^\pi=3/2^-$ of the core nucleus to form the total spin $S=1^-$ or 2^-. The $J^\pi=2^+$ ground state of ^{8}Li or ^{8}B is then constructed by coupling S to the p-wave orbital angular momentum $l=1$:

$$|JM\rangle = \sum_{S=1,2} A_S \sum_{m_l,M_S} \langle l m_l S M_S|JM\rangle |SM_S\rangle Y_{lm_l}(\hat{r}), \tag{7}$$

where A_S are the spectroscopic amplitudes obtained in the shell model. The expression (7) is combined with a common radial wave function for the bound p-wave state. The s-wave capture cross section is then given by the sum

$$\sigma_{rc}(\mathrm{SM}) = \sum_{S=1,2} |A_S|^2 \sigma_S, \tag{8}$$

where σ_S is the cross section associated with the radiative capture into the bound p-wave state from the s-wave channel with total spin S.

The structure of the ^{8}Li or ^{8}B ground state can also be expressed in terms of $p_{1/2}$ and $p_{3/2}$ orbits [12]:

$$|JM\rangle = \sum_{j=1/2,3/2} \theta_j \sum_{m,M_c} \langle jmI_cM_c|JM\rangle |I_cM_c\rangle |jm\rangle. \tag{9}$$

This expression is also combined with a common radial wave function. The two representations, Eqs. (7) and (9), are equivalent [12]. It is instructive, however, to destroy this equivalence by calculating the s-wave radiative capture cross section as the incoherent sum over the two p-wave components of the ground state:

$$\sigma_{rc}(\mathrm{inc}) = (|\theta_{1/2}|^2 + |\theta_{3/2}|^2)\tfrac{1}{2}(\sigma_1 + \sigma_2). \tag{10}$$

The factor $1/2$ arises when each p-wave orbit is decomposed into $S=1$ and 2 components. The label "inc" refers to the incoherent sum. The sum of the spectroscopic factors is the same in the two representations—i.e., $|\theta_{1/2}|^2+|\theta_{3/2}|^2=|A_1|^2+|A_2|^2$—so we obtain the following ratio of the two cross sections, Eqs. (8) and (10):

$$\frac{\sigma_{rc}(\mathrm{SM})}{\sigma_{rc}(\mathrm{inc})} = \frac{|A_1|^2\sigma_1 + |A_2|^2\sigma_2}{(|A_1|^2 + |A_2|^2)\frac{1}{2}(\sigma_1 + \sigma_2)}. \tag{11}$$

We noted earlier that $\sigma_1 \approx \sigma_2$ for the low-energy proton capture on ^{7}Be. This implies that the ratio in Eq. (11) is approximately equal to 1.0, so there is no difference between the two approaches in this case. For the thermal neutron capture on ^{7}Li we found that $\sigma_2 \approx [2.8\pm0.1]\times\sigma_1$. This implies a ratio of 1.24 in Eq. (11) when we insert typical shell model values [11] $|A_1|^2\approx 0.25$ and $|A_2|^2 \approx 0.75$. This explains why the cross sections for neutron capture on ^{7}Li are larger in the shell model approach.

The model that has been used in this work is an example of the incoherent p-wave model discussed above because it is based on a pure $p_{3/2}$ orbit. The argument for using this approximation is that the spectroscopic factor for the $p_{1/2}$ component is very small. Typical values [12] are $|\theta_{1/2}|^2 \approx 0.06$. This argument is not correct because the two p-wave components should act coherently. We are therefore left with the problem of why the full $p_{1/2}+p_{3/2}$ shell model method cannot reproduce the low-energy ^{8}Li and ^{8}B data simultaneously, whereas the model based on a pure $p_{3/2}$ ground state seems to provide a reasonable description. The authors of Ref. [12] argued that it is the two-body potential model that is problematic, because it may not provide a realistic representation of the interior of the nucleus. Future microscopic calculations will hopefully resolve this problem.

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[1] E. G. Adelberger *et al.*, Rev. Mod. Phys. **70**, 1265 (1998).
[2] A. R. Junghans *et al.*, Phys. Rev. C **68**, 065803 (2003).
[3] L. T. Baby *et al.*, Phys. Rev. Lett. **90**, 022501 (2003).
[4] T. Kikuchi *et al.*, Eur. Phys. J. A **3**, 213 (1998).
[5] B. Davids *et al.*, Phys. Rev. Lett. **86**, 2750 (2001).
[6] F. Schümann *et al.*, Phys. Rev. Lett. **90**, 232501 (2003).
[7] L. Trache *et al.*, Phys. Rev. C **67**, 062801 (2003).
[8] L. Koester, K. Knopf, and W. Waschkowski, Z. Phys. A **312**, 81 (1983).
[9] J. E. Lynn, E. T. Jurney, and S. Raman, Phys. Rev. C **44**, 764 (1991).
[10] H. Esbensen and G. F. Bertsch, Nucl. Phys. **A600**, 37 (1996).
[11] F. C. Barker, Aust. J. Phys. **33**, 177 (1980).
[12] B. A. Brown, A. Csoto, and R. Sherr, Nucl. Phys. **A597**, 66 (1996).
[13] H. M. Xu, C. A. Gagliardi, R. E. Tribble, A. M. Mukhamedzhanov, and N. K. Timofeyuk, Phys. Rev. Lett. **73**, 2027 (1994).
[14] Y. Nagai *et al.*, Astrophys. J. **381**, 444 (1991).
[15] J. C. Blackmon *et al.*, Phys. Rev. C **54**, 383 (1996).
[16] J. P. Schiffer, G. C. Morrison, R. H. Siemssen, and B. Zeidman, Phys. Rev. **164**, 1274 (1967).
[17] P. Descouvemont and D. Baye, Nucl. Phys. **A567**, 341 (1994).